

## DE AAR, South Africa

WMO: 685380

Lat: 30.6650S

Lon: 23.9928E

Elev: 1286

StdP: 86.80

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -0.6       | 0.8 | -15.7                         | 1.1 | 15.9 | -13.5 | 1.4 | 15.9 | 14.0                  | 16.0 | 11.6 | 13.2 | 3.5                   | 150  | 0.524 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 14.5                   | 35.0            | 15.6 | 33.8 | 15.6 | 32.7 | 15.4 | 19.9                | 26.6 | 19.1 | 25.9 | 18.5 | 25.5 | 6.6                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 15.3 | 21.5 | 17.4 | 14.6 | 21.2 | 16.6 | 13.8 | 20.9 | 63.2          | 26.5 | 60.4 | 25.5 | 58.2 | 25.5 | 22.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.2              | 10.6 | 9.4 | -3.6                       | 37.4 | 1.2                | 1.3 | -4.5                                               | 38.3 | -5.2       | 39.1 | -5.9       | 39.8 | -6.8       | 40.8 |
|                   |      |     | -4.5                       | 20.7 | 1.2                | 0.9 | -5.4                                               | 21.4 | -6.1       | 21.9 | -6.7       | 22.4 | -7.6       | 23.1 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.5        | 24.1  | 23.9  | 21.6  | 17.2  | 13.6  | 10.1  | 9.9   | 12.0  | 15.5  | 18.6  | 20.5  | 23.1  | (6)  |
|                                                                        | DBStd       | 6.09        | 2.72  | 2.63  | 3.19  | 3.46  | 3.53  | 2.88  | 3.40  | 4.04  | 4.09  | 3.96  | 3.87  | 2.89  | (7)  |
|                                                                        | HDD10.0     | 123         | 0     | 0     | 0     | 1     | 11    | 34    | 45    | 27    | 5     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD18.3     | 1103        | 1     | 1     | 10    | 60    | 148   | 247   | 262   | 199   | 102   | 47    | 23    | 3     | (9)  |
|                                                                        | CDD10.0     | 2851        | 437   | 388   | 361   | 217   | 124   | 37    | 41    | 88    | 170   | 268   | 315   | 404   | (10) |
|                                                                        | CDD18.3     | 790         | 181   | 156   | 112   | 26    | 2     | 0     | 0     | 2     | 17    | 56    | 88    | 149   | (11) |
|                                                                        | CDH23.3     | 9318        | 2098  | 1708  | 1128  | 262   | 33    | 0     | 1     | 37    | 276   | 739   | 1161  | 1876  | (12) |
|                                                                        | CDH26.7     | 3951        | 1014  | 781   | 429   | 47    | 1     | 0     | 0     | 3     | 69    | 257   | 477   | 874   | (13) |
| (14) Wind                                                              | WSAvg       | 4.5         | 5.0   | 4.7   | 4.3   | 3.9   | 3.7   | 3.7   | 4.0   | 4.3   | 4.7   | 5.2   | 5.4   | 5.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 331         | 36    | 51    | 55    | 38    | 17    | 14    | 11    | 14    | 13    | 26    | 30    | 24    | (15) |
|                                                                        | PrecMax     | 666         | 190   | 239   | 166   | 120   | 59    | 87    | 73    | 84    | 87    | 177   | 115   | 119   | (16) |
|                                                                        | PrecMin     | 169         | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |
|                                                                        | PrecStd     | 119         | 38    | 52    | 38    | 28    | 17    | 17    | 18    | 22    | 21    | 32    | 25    | 28    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 37.1  | 35.9  | 34.2  | 30.7  | 26.0  | 21.5  | 22.1  | 26.5  | 31.5  | 33.4  | 34.5  | 36.1  | (19) |
|                                                                        |             | MCWB        | 16.2  | 16.4  | 15.9  | 15.0  | 12.5  | 10.1  | 9.0   | 11.3  | 12.9  | 13.7  | 14.8  | 15.4  | (20) |
|                                                                        | 2%          | DB          | 35.0  | 34.3  | 32.4  | 28.1  | 24.1  | 19.9  | 20.2  | 24.1  | 28.8  | 31.2  | 33.0  | 34.4  | (21) |
|                                                                        |             | MCWB        | 15.9  | 16.4  | 15.7  | 14.6  | 11.8  | 9.3   | 8.9   | 10.2  | 12.3  | 13.7  | 14.5  | 15.1  | (22) |
|                                                                        | 5%          | DB          | 33.6  | 33.0  | 30.9  | 26.5  | 22.6  | 18.6  | 19.0  | 22.5  | 26.6  | 29.6  | 31.5  | 32.9  | (23) |
|                                                                        |             | MCWB        | 15.9  | 16.2  | 15.5  | 14.2  | 11.5  | 8.9   | 8.5   | 9.8   | 11.4  | 13.5  | 14.2  | 15.0  | (24) |
|                                                                        | 10%         | DB          | 32.0  | 31.4  | 29.2  | 24.7  | 20.9  | 17.1  | 17.5  | 20.6  | 24.5  | 27.7  | 29.7  | 31.4  | (25) |
|                                                                        |             | MCWB        | 15.9  | 16.2  | 15.3  | 13.5  | 11.2  | 8.3   | 8.0   | 9.0   | 10.6  | 12.9  | 13.9  | 14.8  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 21.0  | 20.7  | 20.0  | 17.9  | 14.9  | 12.0  | 11.3  | 13.2  | 15.3  | 17.2  | 18.7  | 19.6  | (27) |
|                                                                        |             | MCDB        | 27.6  | 28.5  | 25.6  | 23.5  | 20.0  | 18.3  | 17.7  | 21.5  | 24.7  | 25.4  | 25.0  | 26.7  | (28) |
|                                                                        | 2%          | WB          | 20.0  | 19.8  | 18.8  | 16.7  | 13.6  | 10.7  | 10.0  | 12.0  | 13.9  | 16.1  | 17.4  | 18.4  | (29) |
|                                                                        |             | MCDB        | 26.6  | 26.9  | 25.1  | 23.2  | 19.9  | 16.7  | 17.7  | 20.2  | 23.4  | 25.0  | 24.7  | 25.9  | (30) |
|                                                                        | 5%          | WB          | 19.1  | 19.2  | 18.0  | 15.8  | 12.7  | 9.8   | 9.2   | 10.8  | 12.9  | 15.1  | 16.4  | 17.6  | (31) |
|                                                                        |             | MCDB        | 25.9  | 26.1  | 24.7  | 22.1  | 19.8  | 16.3  | 17.3  | 19.6  | 22.9  | 24.9  | 24.6  | 25.7  | (32) |
|                                                                        | 10%         | WB          | 18.4  | 18.6  | 17.3  | 15.0  | 11.9  | 9.0   | 8.4   | 9.8   | 11.8  | 14.2  | 15.5  | 16.9  | (33) |
|                                                                        |             | MCDB        | 25.7  | 25.5  | 24.1  | 21.2  | 18.8  | 15.7  | 16.4  | 18.8  | 22.0  | 24.4  | 25.0  | 25.4  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 14.5  | 13.8  | 13.2  | 12.2  | 11.8  | 12.1  | 12.7  | 13.6  | 14.5  | 14.9  | 15.0  | 15.1  | (35) |
|                                                                        |             | MCDBR       | 16.5  | 15.6  | 15.1  | 13.8  | 13.5  | 13.7  | 14.1  | 15.2  | 16.4  | 16.3  | 16.7  | 16.7  | (36) |
|                                                                        | 5% WB       | MCWBR       | 4.8   | 4.6   | 4.8   | 5.4   | 6.0   | 6.9   | 7.0   | 7.4   | 7.4   | 6.3   | 5.3   | 5.2   | (37) |
|                                                                        |             | MCWBR       | 12.8  | 12.4  | 12.0  | 11.4  | 11.4  | 11.9  | 12.8  | 13.6  | 14.4  | 14.3  | 14.1  | 13.8  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.315 | 0.323 | 0.313 | 0.290 | 0.269 | 0.255 | 0.255 | 0.293 | 0.319 | 0.326 | 0.300 | 0.302 | (40) |
|                                                                        |             | taud        | 2.550 | 2.522 | 2.529 | 2.583 | 2.626 | 2.636 | 2.624 | 2.475 | 2.374 | 2.395 | 2.540 | 2.574 | (41) |
|                                                                        | Ebn at Noon |             | 1026  | 998   | 976   | 951   | 928   | 921   | 936   | 932   | 952   | 984   | 1034  | 1042  | (42) |
|                                                                        |             | Edh at Noon | 109   | 109   | 101   | 86    | 73    | 68    | 72    | 94    | 116   | 122   | 110   | 107   | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 7.98  | 7.24  | 6.21  | 4.82  | 3.85  | 3.38  | 3.71  | 4.62  | 5.92  | 7.12  | 8.06  | 8.50  | (44) |
|                                                                        | RadStd      |             | 0.41  | 0.40  | 0.27  | 0.22  | 0.16  | 0.19  | 0.15  | 0.24  | 0.29  | 0.22  | 0.34  | 0.40  | (45) |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page